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Optimizing patient transportation by applying cloud computing and big data analysis

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Abstract

In recent years, patient transportation demand has increased rapidly worldwide, especially in the Republic of Korea. Patient transportation requires high accuracy, arrangements, and reasonable allocation in time to the nearest healthcare facilities to support healthcare efficiency. Based on the emergency medical services system process, this research proposes a complete approach for collecting and processing data, and building the operating system, a web-app architecture using ReactJS, NodeJS, and Python to optimize patient transportation based on pathologies, distances, and corresponding specialized hospitals. Our system was designed to crawl data from a public information website of Busan City. After that, it automatically aggregated and stored these data in MongoDB before processing and input into our system. The concept of big data analysis was also built in here. This crawled data were analyzed and applied the API shortest direction of Naver Cloud to identify recommended hospitals with the most intelligent route and least cost. We built a web app connected to a server to visualize the research results and recommend decision-making for the operator and dispatcher in each area. The experiment results showed that the proposed method could recommend the nearest healthcare facilities and routes based on pathologies, optimal distances and times, and travel costs in an actual application, thereby helping solve patient congestion problems, allocate appropriate medical resources, and support healthcare efficiency to solve significant social problems.

Keywords Cloud computing · Big data analysis · Hospital big data · Patient transportation · Virtualization

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1 Introduction

Healthcare systems worldwide are facing challenges related to an increasingly aging population, rates of chronic disease, and new modern but cost-effective forms of treatment. These factors require more intelligent health systems at all levels, i.e., hospitals need to consider solutions and means to increase efficiency and productivity to treat more patients without increasing costs [1]. The trend of Industry 4.0 technology is changing and affecting almost all industries globally, especially the healthcare industry, which is embracing the potential benefits of new innovative technologies such as information systems, environment, health, and management [2]. In the era of Industry 4.0 technology transformation, concepts such as big data, the Internet of things, cloud computing, artificial intelligence, and machine learning are becoming more and more popular. Notably, the picture archiving and communication system (PACS) framework putting away and communicating pictures in medication assists with snapping pictures without printing film, decreasing extra room, recovery time, and sending patient records in clinics [3]. Picture information and patient-related data are effectively obtained by both numerous specialists and specialists on a PC. An electronic clinical record framework and the board of clinical assessment and therapy framework (therapy) [4]; online medical consultation (OMC) [5]; electronic endorsing module connected with the electronic solution entry, etc., have been developed and used popularly in the healthcare system. The need for patient transportation worldwide has expanded in recent years, in relation to medical emergencies, outpatient emergencies, people with disabilities, solitary seniors, or seniors in need of medical care but cannot afford hospital fees. Developing a patient transportation network also helps patients receive more timely and appropriate medical care, with the proportion of patients using outpatient emergency services and admitted to a suitable trauma center being twice as high as those using private means. While it takes less time for a patient to reach a medical facility in a private car, it takes longer for subclinical techniques and emergency care in a hospital. The patient is transported by ambulance without patient information being provided during the transport, making the receiving hospital prepare in advance. There are many organizations responsible for transporting patients from home to medical centers [6]. Sorting and arranging transportation to appropriate healthcare facilities is critical for different patient groups. Transporting a patient from home to a treatment facility or from one facility to another may not require prior onsite emergency treatment and continued resuscitation. Alternatively, medical staff needs to give first aid, then transport the patient to a hospital for further treatment, and continue resuscitation inside the vehicle. Optimal routes for transporting can be an important problem that depends on the characteristics of their location. The main objective of optimal routes is to reduce the amount of time in an ambulance which could prevent a comfortable ride and produce additional health-related problems for patients [7]. The challenges are receiving information from an emergency center and arranging and dividing the optimal route to transport a patient. However, planning and specifying the fastest route to medical

facilities will often take time and labor. Thus, using the direction algorithm and finding the shortest path are expected to help reduce time and increase accuracy. This paper applies the Naver Directions API to provide directions/routes and the expected arrival time, which can predict the time required to travel between branches by reflecting real-time road traffic information time without additional hardware equipment.

This paper presents an optimized patient transportation system based on hospital pathology and distance by applying cloud computing, a web application, ReactJS, NodeJS, and Python. An approach based on methods for building patient, pathology, and hospital database management systems is presented. In addition, procedures for receiving and processing information must be rigorously developed to meet speed, accuracy, and confidentiality criteria. The integrated data is then aggregated and applied to algorithms to return appropriate decisions about choice of routes and hospitals. In order to understand and collect necessary information on patients and pathologies, data models also are appropriately identified in the paper. For example, patient information includes the accident or residential area, background pathology, medical history, onsite foreign or domestic trade, etc. This crawled data are analyzed and applied to the API shortest direction of Naver Cloud to identify recommended hospitals with the most intelligent route and least cost. We have built a web app connected to a server to visualize the research results and recommend decision-making for the operator and dispatcher in each area.

This paper is organized as follows. Chapter 1 introduces the importance of outpatient emergencies, and Chapter 2 provides an overview of previous research or studies related to patient transportation. Then Chapter 3 discusses the methodologies for collecting health facilities data, research design, and process data, building a data model and analyzing the location, while Chapter 4 presents the results and discussion. Finally, Chapter 5 presents the conclusions of this research and future work.

2 Related research

2.1 Health system and patient transportation in the Republic of Korea

According to the Foundation for Research on Equal Opportunity—the World Index of Healthcare Innovation, with the miraculous recovery and quick increase in universal health coverage, the Republic of Korea is well-ranked in terms of quality (12th place), while also ranking third in measuring preventable diseases and top three in patient-centered customer service (11th place) [8, 9]. To achieve this, the Republic of Korea gathered all domestic private health insurance companies into a national health insurance system with an integrated benefits package for people who are insured nationwide despite having different health insurance premiums (systematic salary) in 2000, with the average co-pay of now around 36% [10]. Health systems in the Republic of Korea include national hospitals, specialized public hospitals, regional medical centers, health centers and sub-centers, and primary health posts [11]. In addition, the number of foreigners who come to the Republic of Korea to learn and work has increased in recent years, and local health facilities also provide

more emphasis on opening international clinics to provide better services to foreign communities. As for classification, this system can be classified into three categories and assessed based on the number of specialists and labor medical facilities. The Republic of Korea is one of the top-tier countries with advanced technology and robotics in the world with hospitals and clinics equipped with the most modern medical and software technology. The trend of the Korean emergency medical service system (EMSS) supports the expansion of outpatient emergency medical services from a medical perspective and imports social security and welfare from a social perspective [12]. In regard to the EMSS operating model from the National Emergency Medical Center, this paper does not mention the effectiveness of the model but rather proposes more research methodologies to implement it more effectively. As shown in Fig. 1, the EMSS process involves two stages as follows. In the first pre-hospital phase, accidents were reported and an ambulance was sent, while guidance for emergency care was provided by a 119 emergency service system, manned by trained operators until the ambulance arrived at the scene. Subsequently, the decision was made for hospitals where patients would be transported by sharing information between ambulances and hospitals using a telecommunications system. Care instructions are also provided in route to a hospital. In the second phase in a hospital, examination of treatment that will be provided at the scene followed by sustainable treatment and thorough examination is needed for diagnosis. After that, the decision on hospitalization (ICU, general ward) or an emergency operation

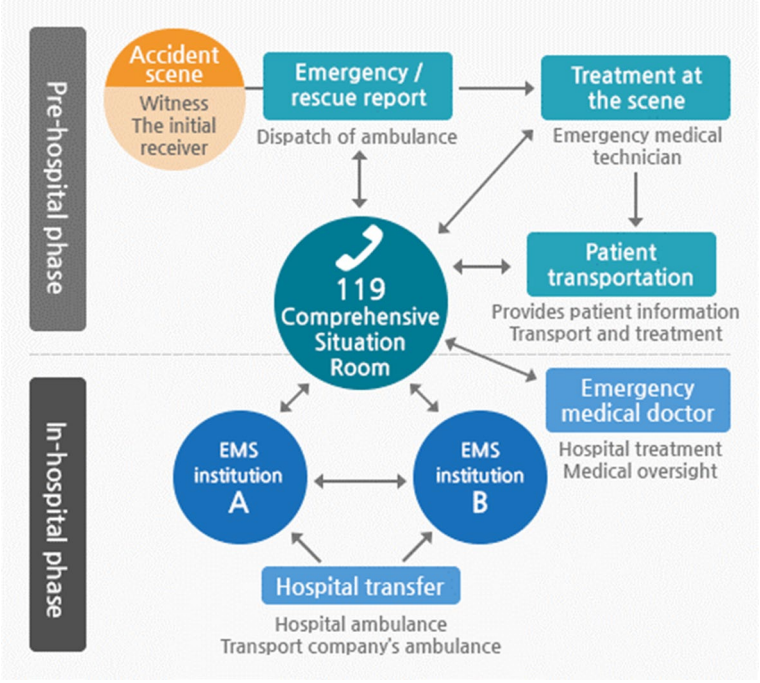


Fig. 1 Process of emergency medical services system (*Source*: www.e-gen.or.kr)

would be made. Finally, it is confirmed whether to move patients to a professional emergency center (for injury, burns, poisoning, cardiovascular center, etc.) equipped with medical teams or facilities and equipment needed for patient emergency care and hospital decisions where patients must be transferred.

In this EMSS process, a shorter time is taken if the cause is diagnosed when the dispatcher retrieves the pathological information, combined with real-time hospital system data to determine the appropriate distance to a hospital. In reality, the ecosystem requires considering many factors, including the data system management of patients, pathology, and hospital information and readiness at the time of the accident. Moreover, the system operates with big data, and aggregate processing in almost real time requires more than simple system architecture. In addition, ensuring that the system operates stably in all disaster scenarios, such as fires, explosions, and earthquakes, in the data center needs special attention because failure to predict these risks will lead to severe consequences, namely sudden stoppage of all emergency services and entire loss of data. Data in the healthcare area, especially emergency medical services, are critical. Thus, in this research, we propose an architecture system with clustering to enhance high availability (HA) and disaster recovery (DR). Using the direction algorithm and finding the shortest path are expected to help reduce time and increase accuracy. In addition, the actual traffic condition en route to medical facilities will also be a factor affecting the best time for patients to be treated. Nowadays, this problem will be solved through technology. In this article, we will provide the solutions.

2.2 Overview of technologies used in supporting patient transportation systems

To improve the quality and efficiency of medical information management, centralized management systems that need to be developed include clinical information, clinical research support, administrative information, management information, education support, and referral systems that operate to generate the best performance when delivering healthcare services [13]. As an example, it is essential to learn and analyze data from hospital information management systems to build a timely monitoring and allocation system of available resources, thereby providing timely instructions for the emergency department. Moreover, many facilities monitor a patient's health status, such as mobile computing, biomedical sensors, and communication devices. With all of these devices, the part of telemedicine monitoring an application communication protocol called WAP (wireless application protocol), could be developed [14]. Owing to the development of medical technology, the average life expectancy of Koreans increased significantly between 1960 and 2006 [15]. The current status of computerization for hospitals allows for making various recommendations to improve the level of computerization and information-sharing between hospitals to strengthen the national health information system [16]. Around the world, developed countries have been applying digital technology to the healthcare system to improve people's lives. A typical methodology adopted and the resulting experience can be found in Granada University hospitals in Spain, which transitioned to complete digital diagnosis [17]. In order for the health system to diagnose

diseases effectively and accurately, it is necessary to seek professional advice from experts. Typically, the diagnostic method is for common and acute pneumonia [18]. In phase with real-time monitoring methods, multidimensional quantitative analysis and patient classification of diagnostic information are done in treating chronic diseases. In addition, home monitoring and treatment methods are integrated to diagnose potential risks [19].

Genetic algorithms were applied in building a non-emergency patient transport service planning system [7]. In this study, from the characteristics of non-urgent patient transport, the arrangement of specific planning for an ambulance given a set of assigned services was researched and built as an experimental model. A genetic algorithm is applied to explore the solution space and includes a scheduling algorithm to generate detailed routes for ambulances. As result, this model can help to decrease the waiting time of patients by up to 13.33 percent. However, the monitoring and adjustment of a route according to the actual conditions have not yet been monitored and adjusted. In our research, we mainly aim for the optimally efficient support model of EMSS service and perform the optimal real-time fastest route. Four approaches, namely Dijkstra, density-based Dijkstra, the evolution strategy, and density-based evolution strategy were performed to reduce emergency services arrival times and avoid traffic jams when there is an accident [20]. A traffic simulation was conducted with fragments of three cities, and the result showed that all approaches could reduce emergency services arrival times by a minimum of 47.9 percent. In this study, using the shortest direction algorithm significantly reduces the travel time of emergency transport when an accident occurs. It is possible when applying the direction-finding algorithm in our integrated system to optimize the EMSS efficiency. Dijkstra's and A^* algorithms are used to calculate the shortest distance from points A (source) to B (destination). The A^* algorithm eliminates fruitless searches by considering the distance to the destination [21]. The A^* algorithm focuses on the destination nodes in order to take parameters, such as time requirement, and distance, then optimizes and chooses the better nodes. Map applications such as Google Maps and Naver Map, also use these algorithms to calculate the shortest path. In this study, with the research context mainly being medical facilities in Busan City, we use Naver Directions API to avail ourselves of the fastest path optimization algorithm that has been built. By using the Naver Directions API, we will also be able to leverage real-time traffic data in Korea.

Healthcare provision is engaged more effectively by inheriting the advantages of cloud computing services. Cloud computing is a service model that allows users to easily access shared computing resources (network, server, storage, applications, services) through network connection anytime, anywhere. The health pilot simulator project in a London hospital was performed to prove the advantages that patients can get from cloud systems [22]. Patients' data storage was built based on cloud computing called the MedCloud system to be presented [23]. Here, the primary concerns were scalability and privacy. This system was deployed with the Hadoop cluster—a big data hardware cluster to enable scalability with the future developing volume of health data. The global navigation satellite system (GNSS) and hybrid cloud computing provide a highly accurate dynamic map for transport or transport applications through leveraging technologies [24]. Cooperative intelligent transport systems

(C-ITS) and cloud-based services were also integrated into an online planning platform. Such connectivity has provided users and public authorities with timely and vital information connections, and will help with transportation issues. Big data start with large-volume, heterogeneous, and autonomous sources with distributed and decentralized control, and seeks to explore complex and evolving relationships among data [25]. In transporting patients, the applications of cloud computing and big data will bring significant benefits and increase the system's efficiency.

Many applications have been developed to facilitate quick and timely access to information to improve the health system, such as applications on mobile phones and websites instead of workstations [26]. Therefore, in this paper, we propose programming techniques for website applications using ReactJS, NodeJS, and Python. Because rapid web technology improvements have allowed software developers to write responsive, mobile-friendly applications quickly, the innovative web frameworks and libraries make it easy to have the same software code base for desktop and mobile devices. The design of the web application can easily be converted into a native mobile application. This allows software products to evolve continuously at a much faster pace with features added on a daily basis [27].

Cloud service (also known as a cloud server) is a virtual infrastructure built to store information processing and applications. A cloud server is a server product similar to a virtual server (VPS server) but is set up with cloud computing technology. Cloud computing is a service that stores data on servers connected to the internet, allowing individuals and businesses to remotely access data anytime, anywhere using a connection network. Cloud computing is being widely used in many businesses and organizations. In the global market, many popular cloud services such as AWS, still lead in global market share for public cloud services at 33 percent, with Azure at 13 percent, and Google Cloud at 6 percent [28]. In Korea, the Naver Cloud Platform has received CSA STAR (cloud information security) certification, which ensures security for cloud service providers under the auspices of the British Standards Institution (BSI) and Cloud Security Alliance (CSA) (USA). Therefore, users can safely process sensitive personal health information on this cloud platform. This cloud service has been used in large-scale national health projects such as Dr. Answer (using artificial intelligence (AI) and big data to guide doctors in diagnosis and patient treatment planning) [29], and the P-HIS (Post, precision, personal health information) system [30]. In this paper, we chose Naver Cloud as our central platform for this study because it is a popular Korean domestic service and can provide numerous solutions and services in the Korean market.

3 Proposal of research model

3.1 Patient transportation optimization model

In our study, user requirements were extracted based on two main methods as follows. In the first stage of our study, we focused on understanding the patient transportation process in hospitals worldwide, primarily in Busan City and other Korean cities. We used the health authority literature, several reports, and this

operational guide as references to get the preliminary user requirements needed for use in the study. Then, we studied the research model and some similar models from previous studies in other countries. We also sought the help of experts in this field to help and evaluate the user requirements in order to get the exact factors that are needed and would work well in the design model in the following stage of the study. In this study, the development of the information system entailed both user requirements and system perspectives. Our proposed model starts with input patient information. Patient information can be received by collecting data in emergency calling and concludes basic information such as identifying a patient's name, age, gender, condition, reasons for calling for help, related symptoms, and mechanism of injury. All this information will be translated into key data by the processing stage. After receiving patient information, we will continue finding optimized routes to match key data with a Korean hospital database and medical treatment status data by applying the "find hospital function" and "find medical treatment status data" in the Python program. With this program, the model can extract all hospital locations in nearby areas by comparing the routes radius. We will then send a request to Naver API Map to retrieve all the real-time paths near the hospital. Here, we have a function to compare and find the best-fit hospital for patients and have the shortest time and routes. We also designed an output map view to allow user visualization. The process and model are presented in Fig. 2.

In this study, with the research context mainly being medical facilities in Busan City, we use Naver Directions API to avail ourselves of the fastest path optimization algorithm that has been built. Using the Naver Directions API, we will also be able to leverage real-time traffic data in Korea.

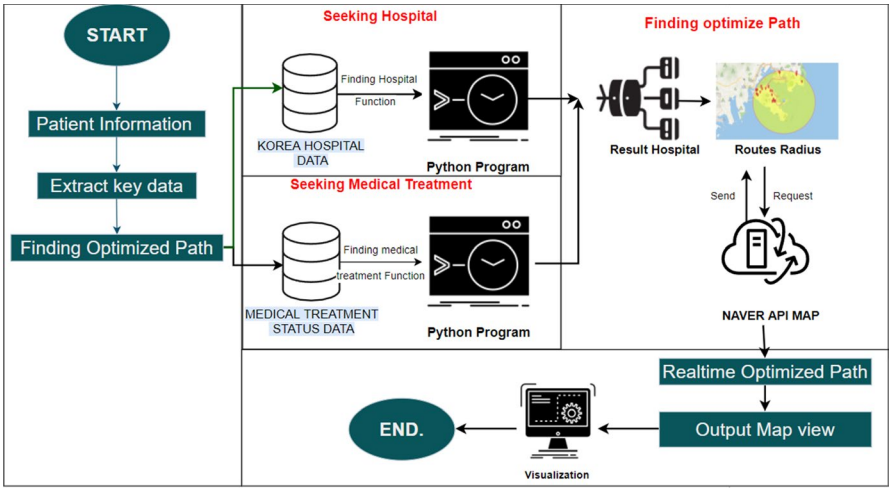


Fig. 2 Patient transportation optimization model

3.2 Proposed operating system framework

In this research, we propose the operating system with three main sections as follows. The first section is the data mining process. We collected data from a public data portal in Korea and aggregated these data using the Python framework, and continued to define the direction of routes information, using an optimized route extraction program applying the API Naver Cloud. We built the database MongoDB to store such information and analysis data. In this research, we chose and used MongoDB as a document-oriented database or known as NoSQL database. MongoDB avoids the relational database's table-based structure to accommodate documents like JSON that have a very flexible schema called BSON. MongoDB uses data stored in document JSON, so that each collection will have different sizes and documents. The data are stored in a JSON document, and the query will be speedy in our case study. MongoDB provides an off-the-shelf, responsive, and extensible platform that allows for breakthrough capabilities, real-time analytics, and global support. Fast querying for real-time analytics on massive user databases with complex data models, flexible schemas, and auto-sharding for horizontal scaling [31] are leveraged. Since it is expected that the system will be designed to access analytical data quickly and accurately, the unstructured data in MongoDB are characterized by no constraints, integrity, availability performance, and easy storage expansion. In addition, data are cached (written buffer) on RAM, limiting access to the hard drive, and so read and write speeds are high.

The following sections show that we built a web-app using ReactJS and NodeJS to visualize and create the user interface. Sections 1 and 2 are connected by a web server based on the Naver Cloud Platform, as discussed in Sect. 3. The research framework is shown in Fig. 3, and the details of each primary process in this research framework are explained in the following sections.

The architecture of the operating system is shown in Fig. 4. We have built our server as a high-availability server model. High-availability server systems continue functioning even after a system hardware or software failure [27]. This means that servers or devices are always in a state of ready service, minimizing the possibility of a system disruption. That is, high availability can be understood as any solution or process, or technology aimed at implementing a guaranteed function for an application, while the database can be accessed 24/7 under all conditions.

Our system's hardware is designed with highly available servers that should be resilient to power outages and hardware failures, including hard disks and network interfaces. Servers are installed with an hourly automatic data synchronization feature to prevent problems related to a power failure or internet connection failure. Our software stack, including the operating system and application, is prepared to handle unexpected failures that could require a system reboot. We also strived to build a data backup mechanism and ensure the integrity of the data throughout the system's life. In addition to a local server system, we have also built and designed a system capable of automatically connecting and synchronizing with a cloud server to ensure the system's speed, stability, and accuracy. We save our data in two clusters (primary one in Asia and secondary one in the USA) to avoid unexpected accidents. The application of clients communicates with operating systems through an

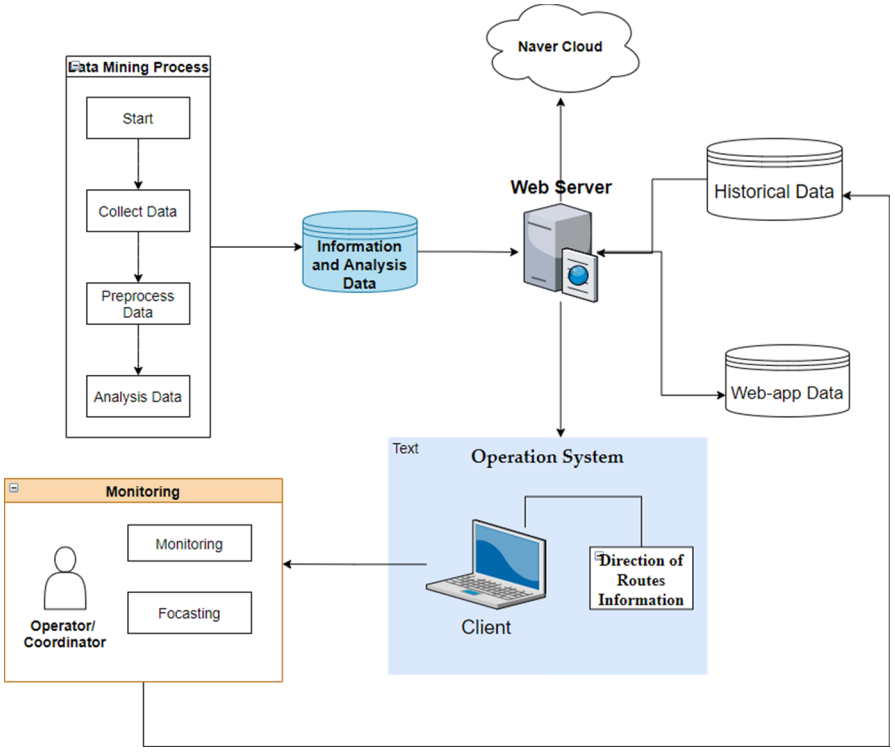


Fig. 3 Research framework

API gateway. Each operation system has a primary server divided into clusters 1 and 2 to avoid data overload. In our operating system infrastructure, the secondary cluster (USA) is defined as adding redundancy to increase reliability and prevent loss of function.

4 Big data processing design and implementation

4.1 Big data collection of Korean hospital and medical treatment status

The classification of pathological information in the preprocessing step will significantly save time for giving solutions. Therefore, big data collection on the specialty information of each hospital from which to build the big data dictionary ultimately and accurately includes the terms "home medicine, internal medicine, anesthesia and pain medicine, pathology, the urology department, gynecology, child and adolescent medicine, neurology, neurology surgery, ophthalmology," etc.

Firstly, information on hospitals is collected from public portals from the Korean government, such as the Korean public big data portal (www.data.go.kr) and Busan Metropolitan portal (https://bigdata.busan.go.kr) shown in Figs. 5 and 6. In this research, we mainly used big data collected from these websites because they are

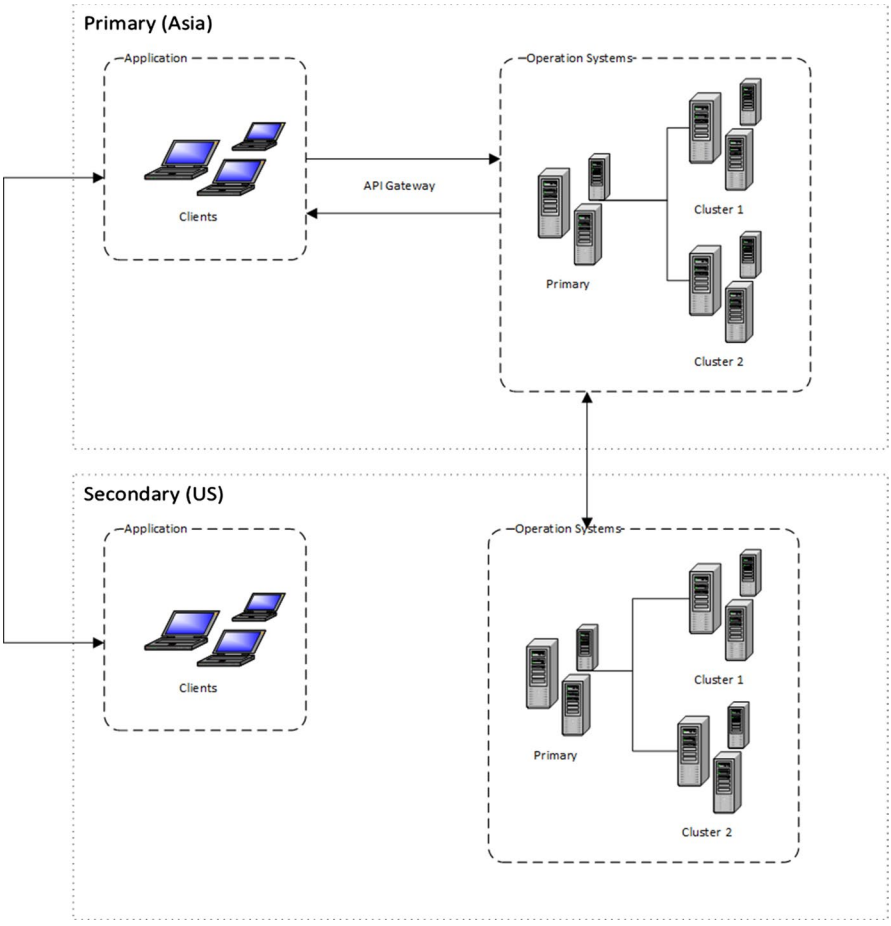


Fig. 4 Operating system architecture

accurate and updated yearly. In general, this is true for hospital big data in Korea, particularly in Busan. The big data that needs to be collected correctly will ensure the efficiency and accuracy of the system later.

In our work, we mainly use secondary data collected from public portals published directly on these websites. Web crawling is an effective method to collect this kind of big data. A web crawler collects data on pages by going from one link to another, and sends that data back to the search engine server. Python combined with Splash-Scrapy is used to crawl hospital information data on the websites. Splash is a lightweight headless browser that works as an HTTP API. We can render JavaScript pages, follow URLs, and extract data from specific HTML paths on each page by using Splash. Our collected data were saved in MongoDB. After collecting data, we cleaned and filtered the data in Python. A key conditions expression was used every time when crawling data to filter data.

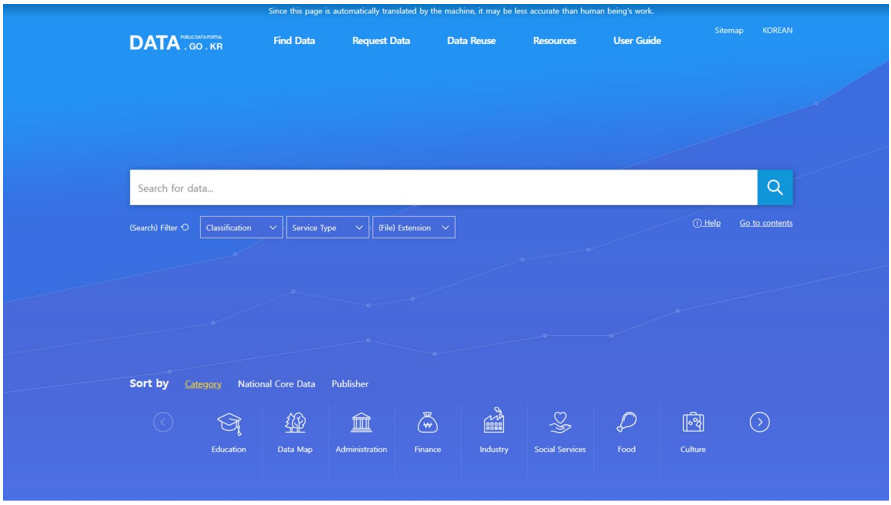


Fig. 5 Korean public big data portal (*Source*: www.data.go.kr)

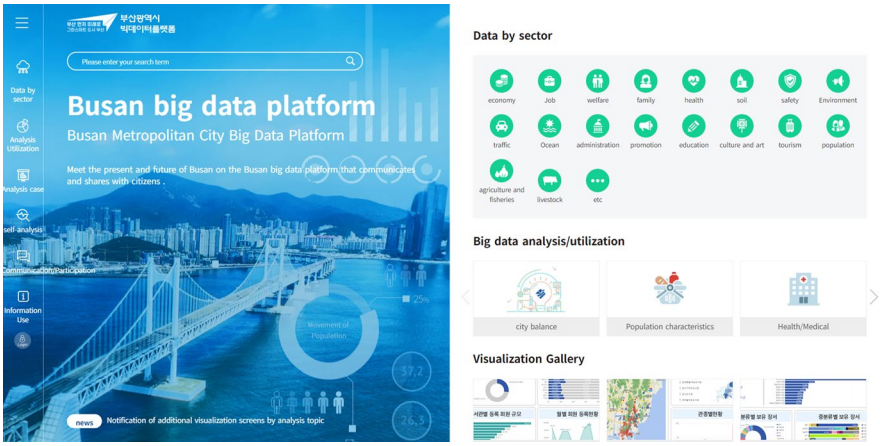


Fig. 6 Busan metropolitan portal (*Source*: bigdata.busan.go.kr)

In the big data of hospital information, key information includes the address, longitude, latitude, medical institution classification, and medical subjects. Firstly, it is necessary to know what kind of medical institution is needed to assist in classifying the severity of a patient's condition to enable saving on treatment costs and human resources at upper-tier hospitals. After that, identifying a patient's condition helps identify the appropriate hospital for a patient's medical condition, shortening the screening time after taking him/her to the hospital. Finally, the address, longitude, and latitude are used to draw up the shortest path from the accident site to a hospital by determining the radius of the accident place from the scene to the identified hospitals and applying navigation tools. The

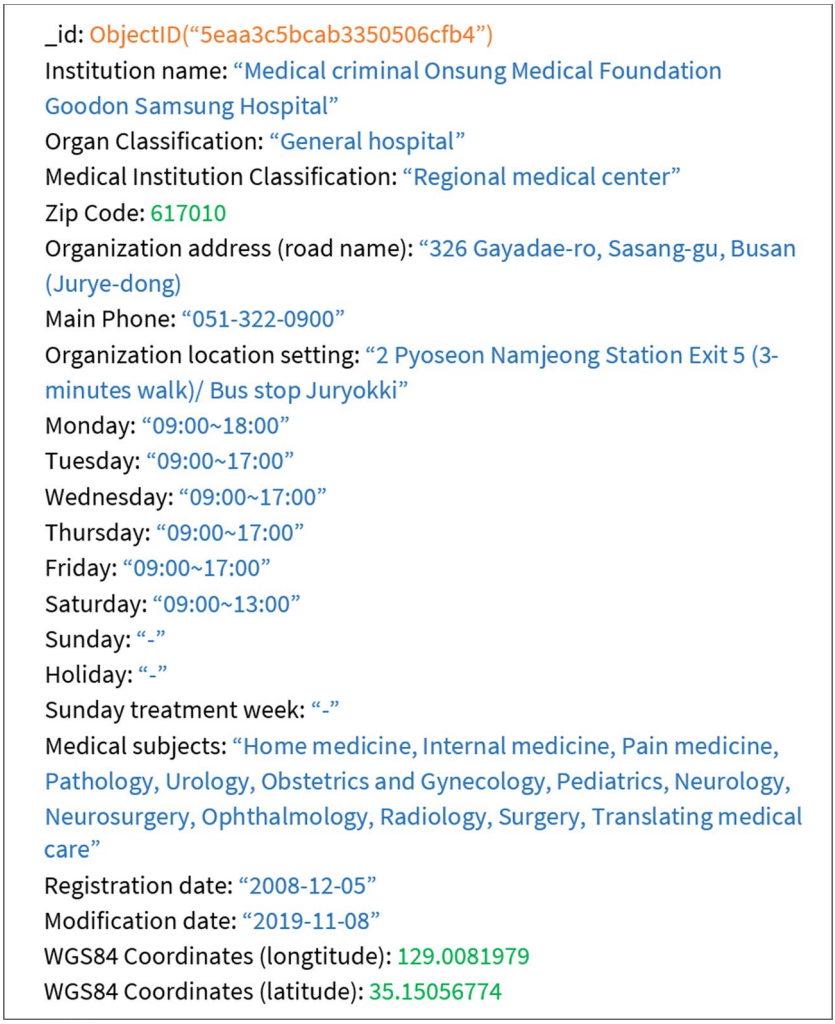


Fig. 7 Databases of specialized hospital

database of metadata on hospital information is shown in Fig. 7, and the user interface is shown in Fig. 8.

Figure 7 presents the structure of specialized big data of hospitals saved in metadata in our system. After processing, specialized hospital big data is stored in MongoDB in the following fields, such as "institution name, organ classification, medical institution classification, zip code, organization address, main phone No., organization location setting, work times (i.e., Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, holidays, Sunday treatment week), medical subject, registration date, modification date, and coordinates" (i.e., longitude, latitude). Each hospital's big data saves the information separately by ID for easier extraction and analysis.

Name	Classification	Medical Subjects	Organ Class	Phone	Address	ZipCode
Eunseong Medical Foundation & Good Samsun Hospital	Regional Medical Center	Department of Family Medicine, Internal Medicine, Middle Medicine Pathology, Vino Medicine, Obstetrics, Gynecology, Pediatrics, Adolescents, Neurosurgery, Oriental Medicine, Imaging Medicine, Department of Surgical Medicine, Injury Operation, Mental Health, Medicine Orthopedics, Diagnostic Examination Medicine, Dentistry, Endodontics	Centre Hospital	051-322-0900	326, Gaya-daero, Sasang-gu, Busan (July Re-dong)	617010
Hakun Oriental Clinic	Emergency Medical Care	Sasang Institution and Acupuncture 2: Oriental Internal Medicine, Oriental Gynecology, Oriental Pediatrics, Oriental Neuropsychiatry, Dermatology, Oriental Medicine	Oriental Clinic	051-514-1075	1943, 2 Jungang-daero, Geumjeong-gu, Busan (Ding by Danbyul, Guseo-dong)	45230
Haein Oriental Clinic	Healthcare Institution	Sasang Quality Department Cheongsam: Department Oriental Internal Medicine, Oriental Gynecology, Oriental Pediatrics, Oriental Neuropsychiatry, Oriental Medicine Dermatology, Oriental Medicine	Oriental Clinic	051-728-7075	11, Banggok 45023-ro, Guyeon, Jeonggwan-myeon, Gijang-gun,	45023
Haein Oriental Clinic	Emergency Medical Care	Sasang Institution: Acupuncture Department, Oriental Internal Medicine, Oriental Medicine, Gynecology, Oriental Medicine, Neuropsychiatry, Oriental Medicine, Neuropsychiatry, Oriental Medicine, Bi-Into Dermatology, Oriental Medicine, Self-sufficiency Medicine	Oriental Clinic	051-714-5125	1.2-floor, 131, Bugok-ro, Geumjeong-gu, Busan (Bugok-dong)	45272
Haein Oriental Clinic	Emergency Medical Care	Sa Wound Pill Chungcheong: Department Oriental Internal Medicine, Gynecology, Pediatrics, Oriental Medicine Sankyung, Psychiatry, Dermatology After Unintentional Medicine	Oriental Clinic	051-861-8675	4, Dongmyeong-ro, Busanjin-gu Busan (Yangjeong-dong)	47212
Haein Oriental Clinic	Emergency Medical Care	Acupuncture Department Oriental Medicin	Oriental Clinic	051-523-9183	20-beon-gil, Dongnae-gu, Busan	47783
Haein Oriental Clinic	Medical Institution	Bedding and herbal medicine: Oriental gynecology, herbal pediatrics, herbal neuropsychiatry, herbal dermatology after nasal cavity	Oriental Clinic	051-201-3232	184 Nakdong-daero, Saha-gu, Busan	49338
Ham Soah Oriental Clinic	Emergency Medical Care	Sasang Nitriding Acupuncture: Testicular Room Internal Medicine, Hospital Gynecology, Hospital Pediatrics, Oriental Neuropsychiatry, Postoperative, Dermatology	Oriental Clinic	051-704-3366	9, Yeongun-le, Haeundae-gu, Busan (Jwa-dong, Daeha Plaza)	48106
Ham Soah Oriental Clinic	Emergency Medical Care	Sasang Jejil Department Acupuncture: Internal Medicine, Gynecology, Oriental Pediatrics, Oriental Neuropsychiatry	Oriental Clinic	051-271-1075	428 Nakdong-daero , Saha-gu, Busan Metropolitan City (Dangni-dong)	49326

Fig. 8 Metadata of hospital information

Figure 8 present the table illustrating the specialized big data of a hospital in the user interface, which can be called metadata of hospital information. The user can search the hospital by name or medical subjects in this table. Specialized hospital big data are arranged by the following columns labeled "institution name, classification, medical subjects, zip code, organ class, main phone No., address," as shown in Fig. 8.

For further analysis, aside from the classification from the hospital attribute big data, medical treatment status information should be mined to assess the hospital's

performance, from which appropriate allocation and management load balancing with hospitals in the same specialized treatment function are enabled. For this big data, the status of the number of beds by function by region and type of care organization is being recorded and analyzed.

The database of metadata on medical treatment status is shown in Fig. 9, and the user interface of medical treatment status is shown in Fig. 10. Figure 9 presents the structure of medical treatment status following separately processing hospital big data and saving the metadata in our system. After processing, the specialized medical treatment status is stored in MongoDB in the following fields of "city, care institution, number of private rooms, number of newborn rooms, number of recovery rooms, number of artificial kidney rooms, number of physical therapy rooms, number of intramural treatment rooms, amount of radiation iodine, and day ward." Each big data saves the information separately by ID for easier extraction and analysis.

Figure 10 presents a table illustrating the medical treatment status in the user interface. The user can search the medical treatment of each hospital by city or

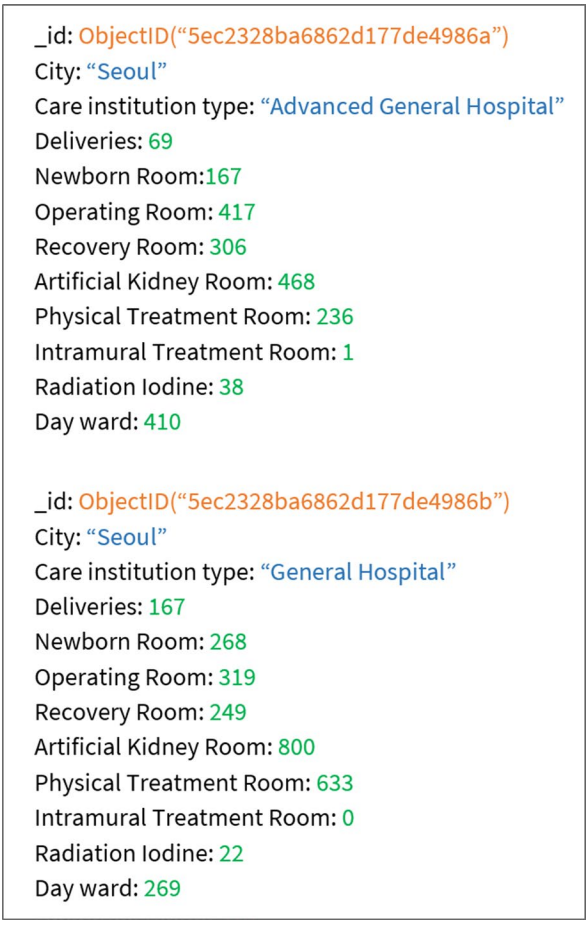


Fig. 9 Databases of medical treatment status

Enter city ... Enter Institution type ...

CITY	INSTITUTION TYPE	PRIVATE ROOM	NEONATAL ROOM	OPERATING ROOM	RECOVERY ROOM	ARTIFICIAL KIDNEY	PHYSICAL THERAPY	DAY WARD
Seoul	Advanced Hospital	69	167	417	306	460	236	410
Seoul	Central Hospital	167	268	319	249	800	800	269
Seoul	Hospital	102	505	575	539	102	2.497	448
Seoul	Nursing Hospital	0	0	0	0	437	1082	90
Seoul	Health Clinic	137	338	858	2.484	1.864	17.543	1.100
Seoul	Dental Hospital	0	0	13	20	0	13	4
Seoul	Dental Clinic Hospital	0	0	0	0	0	105	0
Seoul	Public Health	0	0	0	0	0	15	0
Seoul	Oriental Medicine Hospital	0	0	1	1	0	357	0

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Fig. 10 Metadata of medical treatment status

institution type in this table. The medical treatment status is arranged by the following columns labeled "city, institution type, number of private rooms, number of neonatal rooms, number of operating rooms, number of recovery rooms, number of artificial kidney rooms, number of physical therapy rooms, number of intramural treatment rooms, amount of radiation iodine, and day ward."

After that, collected data are stored and designed in MongoDB, an object-oriented, simple, flexible, and scalable NoSQL database, and is based on the NoSQL document storage model. Big data objects are stored as separate documents within a collection instead of storing in traditional relational database columns and rows. The MongoDB language deploys a big data warehouse that provides high performance, high availability, and auto-expansion. MongoDB uses JSON or BSON documents to store big data.

4.2 Big data processing and mining

For big data processing and mining, the Python language is used in this study. As an open-source programming language developed through a community-based model, Python is widely used for computer science in academia and applied in many industries. Python has many well-tested analytical libraries that include numerical computing, big data analysis, statistical analysis, visualization, and machine learning. As Python is a high-level language, it has many advantages to help speed up projects, and allows sketching ideas to realize them quickly, while ensuring legibility. As an object-oriented programming language (OOP), Python also supports advanced big data structures such as lists, sets, tuples, and dictionaries, as well as many scientific computational activities such as matrices and big data frames. In this study, Python was used to retrieve big data from MongoDB and structured into a data frame of the Pandas library. Since Pandas is open source and well-rated by the community for high-performance, big data processing calculations will be much easier than

traditional ones. The flask, flask_cors, and requests libraries were then used to build an application programming interface in Python.

After analyzing and evaluating the classification of medical institutions and medical subjects, route-oriented information big data is important and necessary to guide the emergency agency in transporting patients to the identified hospital in a timely, accurate manner. In order to have enough conditions to build the route, we need to determine the geographical location by ascertaining a longitude and latitude. Once geographic location parameters are available, determining the distance and radius becomes easier. The route orientation recommendation is shown in Fig. 11.

In processing any big data, big data of hospital information and medical treatment status are structured and statistical. Since raw big data collected from the portals is not eligible for conducting statistical and analytical functions, they require additional filtering and transformation, such as streamlining big data and providing measures. The example (as shown in Figs. 11 and 12) depicts retrieving big data from MongoDB and storing it in dataframe df_hospital_information. After that, big data of medical subjects are extracted and then formatted as an array, so the split function is used to transform big data and store it in another dictionary with two keys as label and value. Nonetheless, based on the frequency of the label's appearance, the label's value will be updated. Therefore, the medical array used to check the frequency and value will be increased by one if the label appears.

Figure 12 presents our extract and transform big data algorithm. Firstly, we created 3 big datasets as ds1, ds2, ds3 of "information hospital," the big dataset to be selected after transforming, and the big dataset for the number of specialty hospitals. To process ds1, we started with the assigned array medical_array, big data frame



Fig. 11 Routes orientation recommendation

```
ds1 = data set of information hospitals;
ds2 = data set to be selected after transformed;
ds3 = data set of number specialty of the hospital;
medical_subject = specialty of the hospital, is the string of words separated by commas ;
Processed(ds1){
        medical_array = empty array;
        df_medical = empty dataframe;
        medical_subjects = Unique(ds1[medical_subject]);
        for subject in medical_subjects:
                for i in Split(subject by comma):
                        if i in medical_array:
                n = CurrentValue(df_medical[i])
                                SetValue(df_medical[i], n + 1)

                        else:
                Append (i, medical_array);
                                Append (dict(key: i, value: 1), df_medical);

        df_medical[% of medical_subject] = value of each label / total value in df_medical;
        ds2 = Selection(df_medical)
        ds3 = Length(Array(Split(subject in medical_subects by comma)));
        return ds2, ds3;
```

Fig. 12 Pseudocode of phase extract and transform big data

df_medical, and variable medical_subject (which is unique medical_subject in ds1). For each subject in medical_subjects, we had a loop for i in delineating subjects with a comma; for i in medical_array, n would be the current value of df_medical[i] and set value as df_medical[i], with n equal to $n+1$, else append value i in medical _array together with append dict with key i and value 1 in df_medical. After that, we could calculate the ratio of each medical subject by the value of each label by dividing the total value in df_medical. We also have ds2 as the selection of df_medical and ds3 as the length of separating medical_subject delineated by a comma, with return values of ds2 and ds3.

4.3 Operating system design and flow processing

The web application is designed in two roles. Firstly, portal management is for the information manager, user privileges, and metadata used to support emergency departments and hospitals, supporting the evaluation of big data as to their accuracy,

and it can be used to monitor the basic information of a hospital from which to provide support strategies as needed. Secondly, emergency management is for the service user, helping determine the route and hospital that suit a patient's pathology. The hierarchy diagram is shown in Fig. 13.

Figure 14 presents the flowchart of the central operating system. The processing flow for the application built in this paper includes the following main components. Firstly, the login function has two main roles of "admin" and "emergency_admin." These two roles are responsible for information security depending on who's allowed to access it. For an "admin," permission is given to access metadata on hospital information and medical treatment status. The task of an "admin" is to build the necessary metadata in support of big data system analysis, optimize the classification and evaluation of load balancing among hospitals in the same institution's medical classification, and provide statistics and analysis to help an emergency agency as much as possible in transporting patients promptly. For the remaining "admin_emergency" permission to enter a patient's pathology, the address of the scene and radius depend on the severity of a patient's condition. The system will recommend the appropriate hospital and determine the route to get there. The map in Fig. 14 shows the directions and information for hospitals in the same medical subjects within a defined radius.

Figure 15 shows the entire UML framework of our operating system. A user can access the system and start with login interface by logging in with a username and password. He/she can also log out and register, and the entire session will be saved in the system.

When a user logs in, the information will connect with the cloud server through the internet network and API. The server also has user, real-time, analysis, and visualization data as well as metadata. After login, the server will return user information. If user information is emergency management, the user will be directed to the "Emergency management" module. Upon receiving patient information, he/she can write and get information, using the "get medical subject, institution type, address

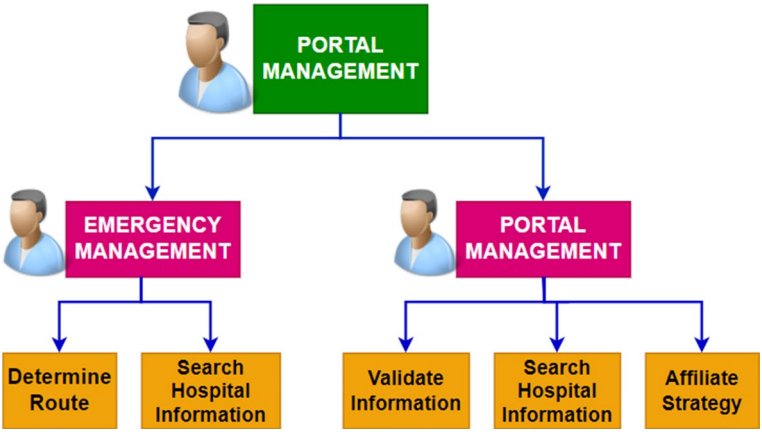


Fig. 13 Hierarchy diagram of role management

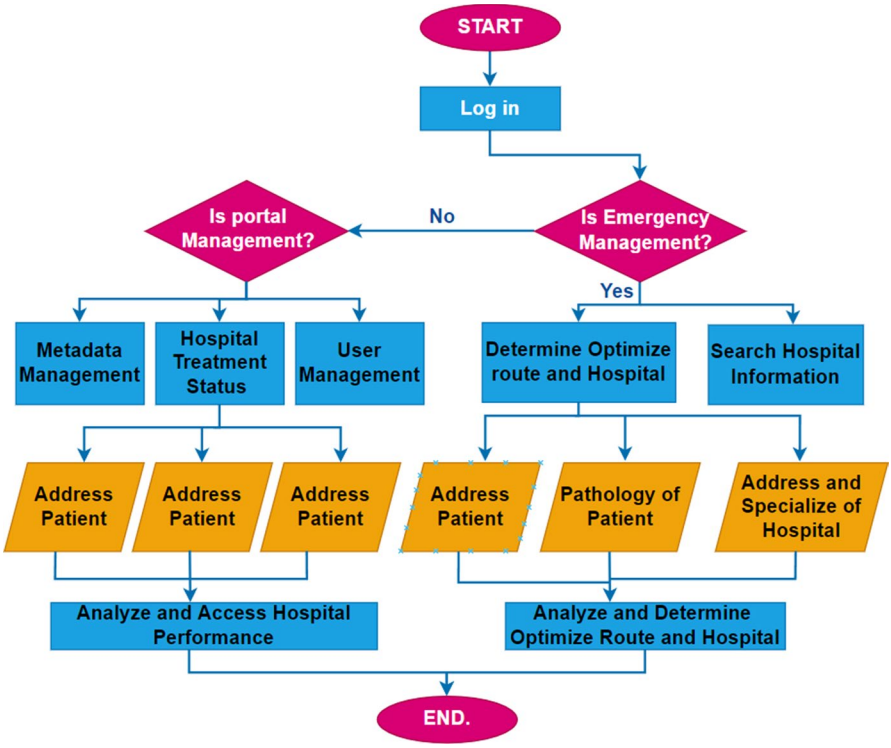


Fig. 14 Flowchart of operating system

patient, and pathology patient" and "search hospital information and optimize analyze" functions. The system will send this information to the "Optimize emergency" module, which in turn will determine the route and hospital using the "isMatchMedicalSubject," "isMatchInstitutionType" and "isShortDistance" functions. After this phase, the system will execute the "goMain" function to go to "Main," which will suggest the route and hospital using the functions "showRoute" and "showHospital." Otherwise, if the user is portal management, the system will send him/her to the "Portal management" module. Portal management allows the user to manage or edit user data, metadata, or the medical subject, as well as institution type in order to modify and manage hospital treatment status and metadata.

4.4 Connect to cloud server

In addition to hospital information and medical treatment big data, the geographical information of an address will be processed through the Naver Cloud Platform as the service application. We used "naver-maps-api" of the application services of the Naver Cloud Platform. Figure 16 presents the process of registering a new "naver-maps-api," and we queried params with three main keys of "start, goal, and option" with a unique value. Finally, "client ID and client secret" are issued to gain access.

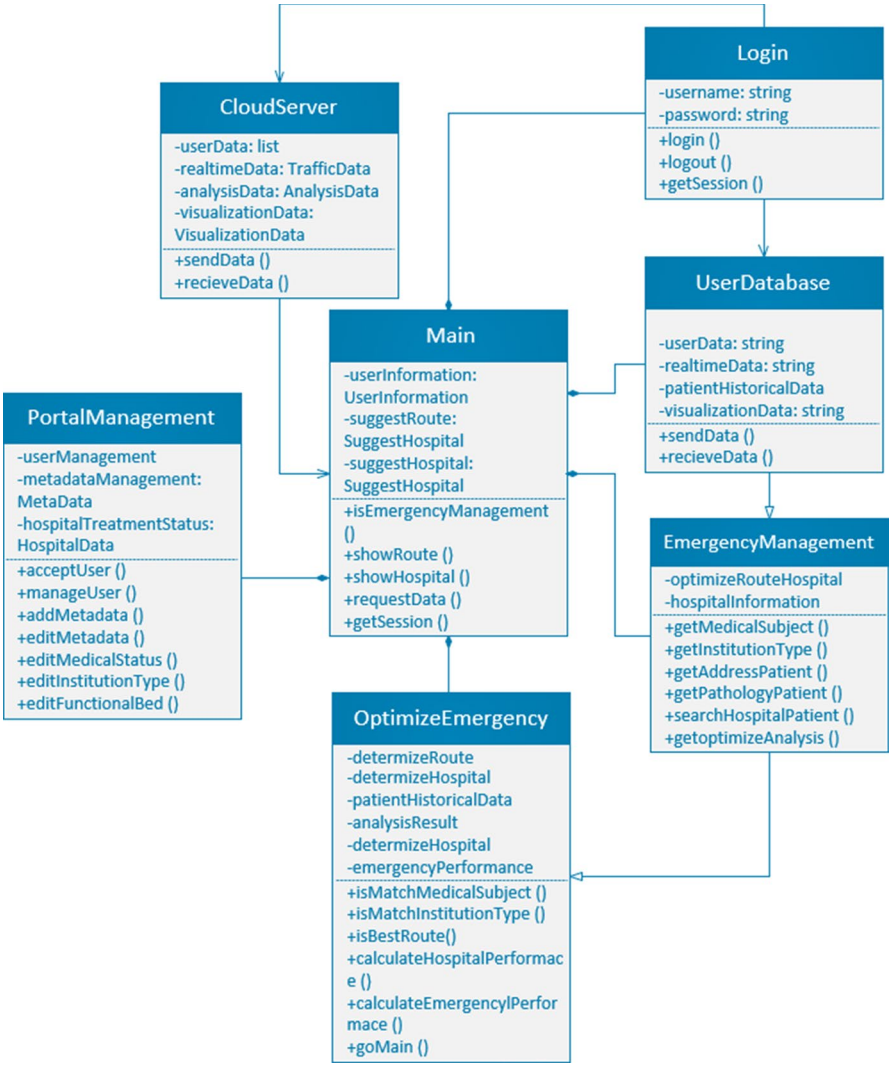


Fig. 15 UML of the main operating system

After registering the service on the Naver Cloud Platform successfully, geocoding and direction information will be provided when sending API requests.

5 Experimental setup and results

The result of this research is the operating system. In this part, we will clearly define the user interface of "web-app."

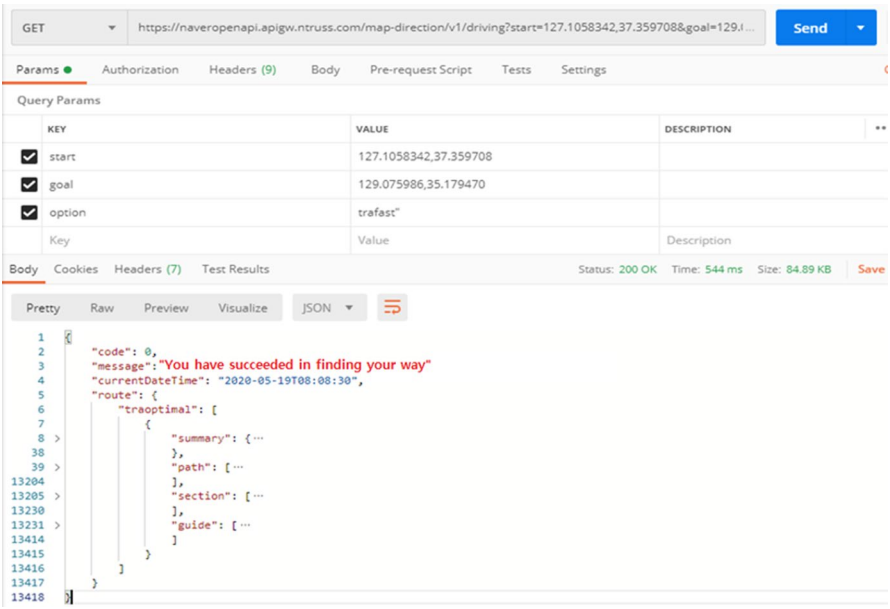


Fig. 16 Instruction of maps services

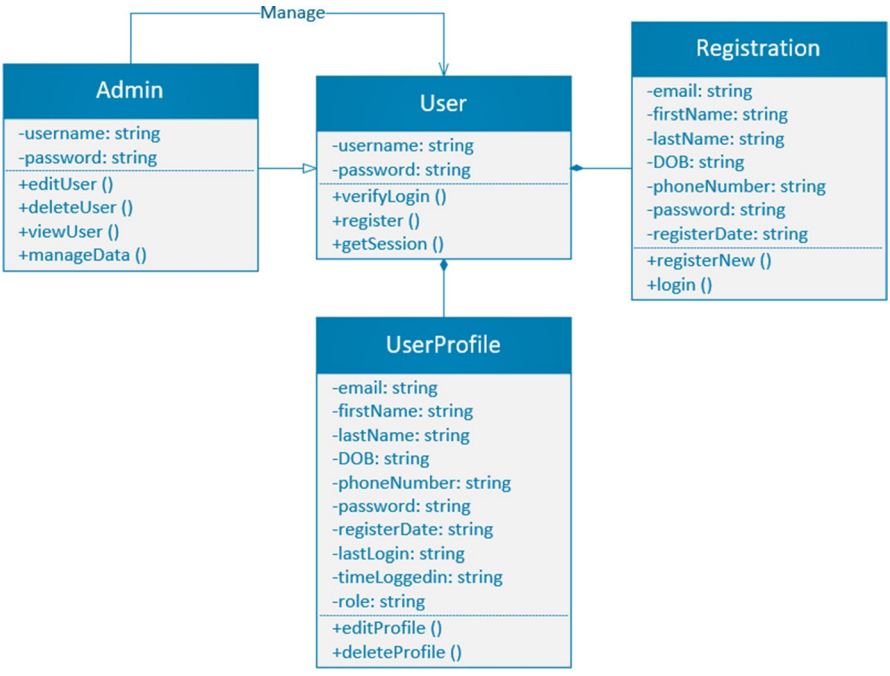


Fig. 17 UML login page

First of all, the UML login page is presented in Fig. 17. Upon user login with a username or password, the system can recognize whether the user is an admin or not. The user can log in, log out, register, and use the "get session" function to record all login sessions. If the user chooses the "register" function in "User," the system will proceed to "Registration," which contains "add firstname, lastname, birthday, email, phoneNumber, password, and role" with the "Register()" function. After "login," if the system recognizes that user_id is admin, the user can use the "Login(), Managedata(), Manageusers(), and Managepage()" functions. Both an admin and normal user can proceed to "Profile," which enables viewing the user's information (i.e., firstname, lastname, birthday, email, phonenumber, password, and role) and using the "edit profile" function.

Figure 18 presents our login and register pseudocode. With a login pseudocode, the user imports his/her username and password. The system will get this username and password, and then the condition clause is used if the username and password are equal to the ones in the user database, and the system will send a "Login successful" notice. Otherwise, the system will send a "Username and password are invalid; please try again" notice. With a register pseudocode, a user will create and import the email, username, and password. We also have "check" function for unique username and email in order to check whether a username is the same as the one in the database. If so, the system will ask the user to change his/her username. We use the condition clause to check if the email, username, and password are blank, and then the system sends a notification "You have not entered enough information; please

```
//Login Pseudocode
If Click Login button {
        GET username;
        GET password;
        IF (username == Search(userdatabase(username)) && Password == Search(userdatabase(password) THEN
                Write ("Login Successful)
        ELSE
                Write("Username or password was invalid. Please try again")
//Register Pseudocode
ELSE
        new email = email of new user, placeholder: "example@email.com": (Function Check Unique username)
        new username = username of new user: (Function Check Unique username)
        new password = password of new user : "more 6 digits"
        Signup Submit = "button submit"
        IF email, username, password is blank THEN
                Write ("You have not entered enough information. Please try again"
                ELSE
                        Write ("Enable Submit to Register")
                        Create (user_id, username, password)
        return user_id, username, password;
```

Fig. 18 Users' login and register pseudocode

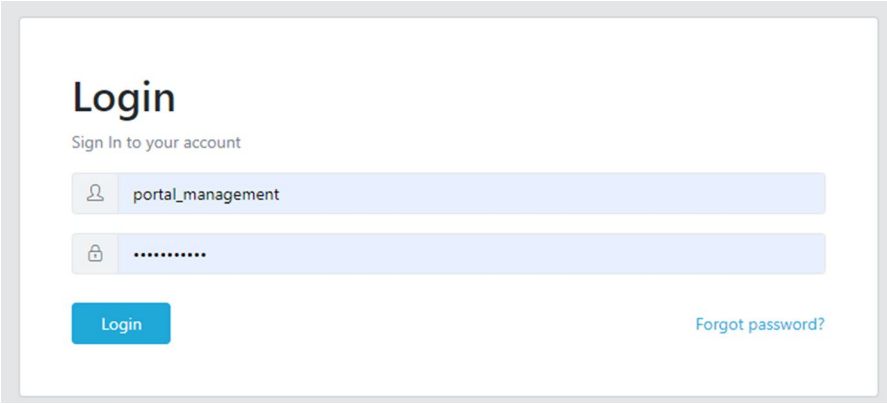


Fig. 19 Login page

try again." Otherwise, the user is able to register. Then the system will create a user_id, username, and password and save them in the database.

Figure 19 presents the main login user interface. A user only needs to submit the username and password that were registered. If he/she cannot remember the password or username, he/she can use the "forgot password" function to recover his/her username or password.

The admin role allows administering users and accessing hospital information and medical treatment status plus analytical reports to assess the situation of load balancing among hospitals in the same medical subjects. Admin will be the role with the most functions. An admin can manage all the accounts saved in the system, and view the details of any user accounts, status, activities, and performances as shown in Fig. 20.

An admin also has permission to follow the metadata of the system, which can include "hospital information, routes, patient information, and historical data" as presented in Fig. 21. In the metadata interface, an admin can access all metadata in the database, which contains the names of hospitals or clinics, classifications, medical subjects, organ classes, phone No., addresses, and zip codes and input these items to search for a specific hospitals or clinics, as well as modify hospital information or add more big data.

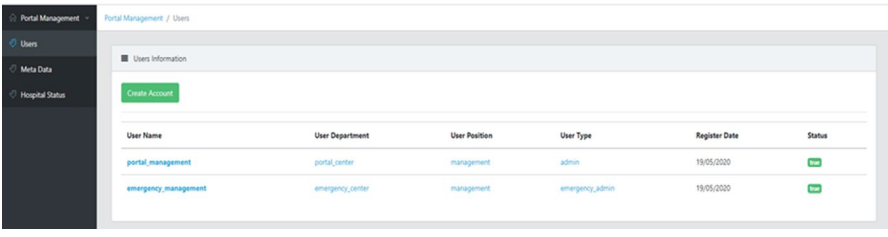


Fig. 20 'User management' portal for admin

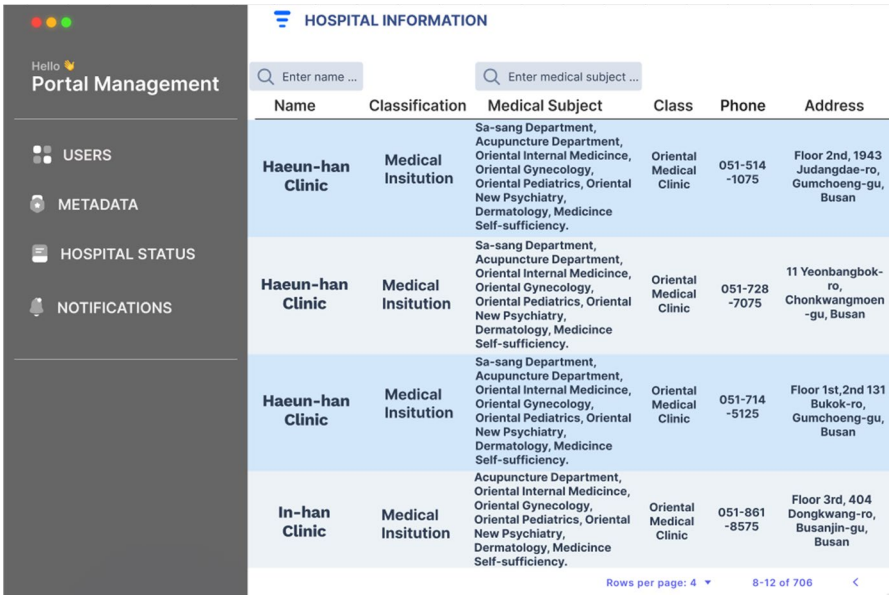


Fig. 21 "Metadata" portal for admin

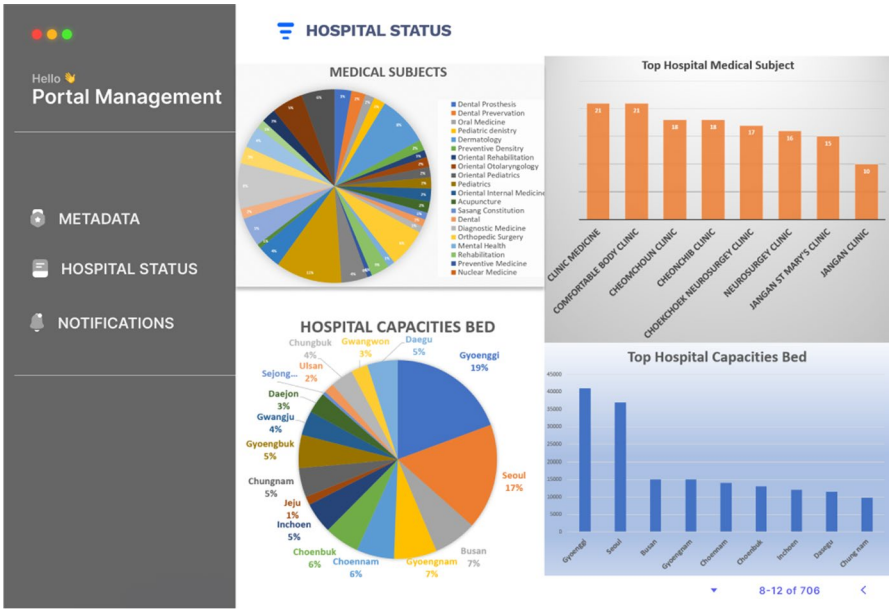


Fig. 22 "Hospital medical treatment status" portal for admin

An admin can see and access all analysis dashboards about the hospital medical treatment status and can also choose what will be displayed in the other main user interface. Figure 22 shows the page hospital medical treatment status for the admin role. In this research, we presented four main types of analysis dashboards. The medical subjects dashboard is presented for the ratio of medical subjects in metadata. There are the top 10 medical subjects, hospital bed capacities, and top 10 bed-capacity hospitals dashboards are available as well.

For an emergency admin role allowed to input the address of the scene, the radius belongs to critical conditions with regard to the severity of a patient's condition and patient pathology for hospital classification. After that, the appropriate hospital listings will be calculated and displayed by the system.

Emergency agents can select the appropriate hospital with the map directions displayed as shown in Figs. 23 and 24. When a user submits the address of a patient as the one listed in Fig. 23, such as "727 Taejong-ro, Youngdo-gu, Busan," the system will provide a list of the names of hospitals or healthcare clinics matching this address. The system displays the hospital names, classifications, medical subjects, organ classes, phone No., addresses, and zip codes. A user can also search by radius (by kilometer) or pathology of patients.

When the user chooses the best-fit hospital or healthcare clinic, the operating system will also provide the directions to this hospital or healthcare clinic. Figure 24 presents a map on the "Intelligent Directions Assistant" emergency portal.

The experiment results pointed out that the proposed method can recommend the nearest healthcare facilities and routes based on pathologies, optimal distances and times, and travel costs in an actual application. It will help solve patient congestion



Fig. 23 "Intelligent directions assistant" emergency portal

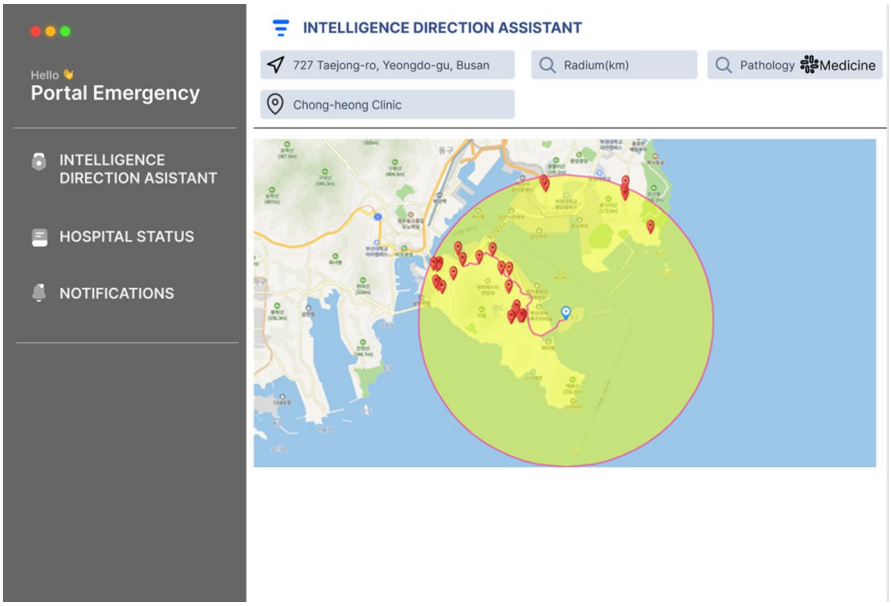


Fig. 24 Map in "intelligent directions assistant" emergency portal

problems, allocate appropriate medical resources, and support healthcare efficiency to solve significant social problems. When users input their current address and pathology, the system displays a list of hospital names, classifications, medical subjects, organ classes, phone No., addresses, and zip codes. The user can search by radius (by kilometer) or pathology of patients. As a comparison, a non-emergency patient transport service planning system was designed applying genetic algorithms [7]. A genetic algorithm is applied to explore the solution space that included a planning algorithm to generate detailed routes for ambulances. As a result, the model can help reduce patient wait times by up to 13.33 percent. However, the characteristics of non-emergency patient transport, specific planning arrangements for ambulances in a range of designated services, were studied and established as experimental models. A traffic simulation was conducted with fragments of three cities, and the result showed that all approaches could reduce emergency services arrival time by a minimum of 47.9% [20]. This studied performed to reduce emergency services arrival times and avoid traffic jams when there is an accident. In our research, we mainly aim for the optimally efficient support model of EMSS service, and perform the optimal real-time fastest route. In this study, with the research context mainly being medical facilities in Busan City, we use Naver Directions API to avail ourselves of the fastest path optimization algorithm that has been built. By using the Naver Directions API, we will also be able to leverage real-time traffic data in Korea. The results of our experiment also showed that by using the above method combination, optimization of the navigation was demonstrated. As soon as a patient's information is entered, the system is able to accurately extract and analyze the results.

6 Conclusion and future work

This research aims to build an optimized patient transportation system based on hospital pathology distance by applying cloud computing and statistical big data visualization to provide medical staff with the necessary information on any hospital and state of medical treatment. Specifically, we crawled medical facilities data of Busan City, including the addresses, longitudes, latitudes, medical institution classifications, and medical subjects. The integrated data are then aggregated and applied to algorithms to inform appropriate decisions as to the choice of routes and hospitals. As a result, the application assists in classifying hospitals according to patient pathologies and route navigation in a timely, effective manner. In this paper, Naver Directions API is applied to provide directions/routes and the expected arrival time, which can predict the time required to travel between branches by reflecting real-time road traffic information time without additional hardware equipment. The experiment results pointed out that the proposed method can recommend the nearest healthcare facilities and routes based on pathologies, optimal distances and times, and travel costs in an actual application. It will help solve patient congestion problems, allocate appropriate medical resources, and support healthcare efficiency to solve significant social problems. This current research needs to be evaluated based on utility and cross-validation metrics with different training and test sets. Building cross-validation and evaluation will be researched more in a future work.

In addition, more patient transportation options have been provided. As an example, a helicopter is a means of transportation that helps physicians reach patients quickly, promptly bringing patients from remote or isolated locales of a city to specialized hospitals in case of an emergency to get prompt treatment. Watercrafts are used to transport patients in need of first aid on rivers to adapt to a localized model with many rivers and canals to overcome the obstacle of traffic congestions. Another possibility is air transport in an emergency because this is an optimal solution for sea-borne rescue operations from an island or southern provinces in an accident needing emergency care. We will research more on the actual demand for these transport patients. The testing method of the research model and evaluation of the sampling procedure is yet to be established. In a future work, more big data will be gathered on patient transport history and real-time big data from information-sharing between hospitals to improve the process load balancing and offer a more efficient navigation assistant. For cost savings and increased processor performance, the entire application architecture and system will be deployed on a cloud platform.

Data availability Please contact the corresponding author for data requests. The languages used were Java, XML, Python, Jason, etc., and they were developed using the JVM (Java Virtual Machine), GVM (Geoblue Virtual Machine), and middleware programs. The database was also stored in the cloud using RDB.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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